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## Introduction

Bar Louie, a once-popular national gastropub chain, filed for bankruptcy after shuttering 17 "underperforming" locations across the U.S. Poor site selection was cited as a major contributor to these closures, showing that even well-known brands can fail when they overlook the importance of location.

Location plays a pivotal role in determining success in the competitive landscape of the restaurant and bar industry. Opening a new business involves strategic decision-making, especially in a competitive city like Philadelphia. How can a prospective bar owner choose the right neighborhood to maximize customer satisfaction, business potential, and yield the best return? Thus framing our core question for prospective bar and restaurant owners: Where should a new bar in Philadelphia be located for optimal visibility, customer satisfaction, and performance?

Platforms like Yelp are becoming a primary source of information for potential customers, understanding what drives positive reviews and high foot traffic is crucial before heading over to a restaurant. By analyzing Yelp reviews, ratings, and geographic data, our goal is to identify "hotspots" in Philadelphia where bars tend to perform well and understand the characteristics, such as bar type, customer demographics, and neighborhood traits, that contribute to their popularity. These insights can help future restaurateurs make data-informed decisions about where to establish a successful bar and what features are most likely to make a bar thrive in a given location.

By using data analysis tools on data acquired through Yelp's API and the Census Geo-Coding database, we will analyze patterns in customer preferences and identify if specific locations contribute to Yelp ratings. We will process and analyze large datasets to uncover the attributes that contribute to restaurant success on Yelp. This exploration could provide useful insights for individuals looking to open a new bar in Philadelphia.

## Data sources and techniques

We used a combination of Yelp's public dataset and U.S. Census demographic data to explore how location and business characteristics impact Yelp ratings for bars in Philadelphia.

Data Sources:

1. Yelp Open Dataset (source):
   Business.json: Contains metadata for each business (e.g., business_id, name, address, stars, categories, attributes, etc.).
   Review.json: Contains individual user reviews (review_id, stars, text, date, useful, etc.).
2. U.S. Census Tract Data for Pennsylvania:
   DP03.csv: Socioeconomic data (e.g., employment status, income).
   DP05.csv: Demographic data (e.g., gender, age).
3. Census Geocoder API:
   Used to generate 15-digit GEOIDs from bar coordinates for spatial analysis.

Data Properties:

- Final dataset: ~1,400 bars filtered to Philadelphia with matched GEOID data.
- Review dataset: 1M+ reviews filtered and joined by business_id.
- Census data includes ~2,500 Pennsylvania tracts with employment, income, age, and gender breakdowns.

Variables:

- Business attributes: outdoor_seating, wifi, has_parking, happy_hour, ambience
- Review metrics: stars, useful, cool
- Census indicators: income_benefits, Above21, employment_status

Data Analysis Strategy:

To analyze the relationship between bar success and location, we first cleaned and standardized our datasets by handling missing values in key fields such as parking, business hours, and categorical attributes such as city and cuisine type. We filtered the data to retain only bars located in Philadelphia and excluded entries with all-zero values or unrelated categories. Business and review datasets were joined using business_id, and geographic identifiers (GEOIDs) were appended using the U.S. Census Geocoder API based on latitude and longitude. We then merged this enriched dataset with tract-level census data to incorporate demographic and socioeconomic variables. Key features such as average ratings, bar counts, and amenity presence were aggregated by GEOID to facilitate tract-level comparisons. This structured pipeline enabled location-based insights while preserving bar-level granularity for further modeling and visualization.

Techniques Applied:

- Exploratory Data Analysis: Bar plots, scatter plots, histograms to uncover trends.
- Feature Engineering: Grouping by GEOID, computing density, binary attribute indicators.
- Topic Modeling: Applied BERTopic on review text to extract key themes.
- Geospatial Analysis: Used Tableau to map star ratings, bar counts, and feature availability by tract.
- Validation: Spot-checked joins, visually confirmed tract boundaries in Tableau, and manually reviewed samples from extreme rating clusters.

## Results

### Identifying Key Drivers of Bars' Star Ratings Using Topic Modeling

To understand what makes certain bars more successful than others, we used BERTopic to cluster over 200000 customer reviews into interpretable topics. Success was evaluated with two outcome variables: review frequency (serving as a proxy for popularity) and average star rating (as a proxy for satisfaction). The distribution of star ratings amongst all topics showed that the majority of the bar-related themes received moderately favorable reviews, with most of them falling between 3.5 and 4.2 stars as seen in Figure A1. However, this overall pattern conceals important variation when we look into specific types of bars more closely.

As witnessed in Figure A3, most topics came out as both highly frequent and revealing in terms of customer satisfaction. For instance, the "cheesesteak" topic was the most frequently mentioned, but it obtained only middling average ratings. This suggests that bars associated with extensively recognized food items such as cheesesteaks may suffer from oversaturation and higher expectations, where strong visibility does not translate into consistently good customer experiences. In contrast, themes revolving around service and ambiance such as "atmosphere, service, food, staff, excellent" were both common as well as well-rated, underscoring that strong execution of the basics continues to encourage positive customer experiences. Notably, commonly occurring food-forward themes like "sushi, quality" and "vegan, vegetarian, options, tofu" also had high average ratings, suggesting that even popular categories can do well when paired with a clear culinary identity and commendable quality. Figure A4 further reinforces this, visualizing that topics like "mussels, monk, frites," sushi, and vegan themes had particularly large proportions of 4- and 5-star reviews.

An evaluation of the top and bottom-rated topics (see Figures A5 and A6) provides added support for this finding. Reviews referencing scents and candles had relatively low review counts but stood out with ideal ratings. This reinforces the idea that bars offering distinctive, curated experiences can generate high satisfaction even without extensive exposure, likely because of their clear identity and consistent delivery. In contrast, low-rated topics often consisted of keywords like "rooftop," "skybar," or "ipad ordering," suggesting that novelty, flashiness, or tech-driven concepts may attract attention but fail to deliver on key service expectations. This is further enforced by the weak overall correlation between topic frequency and average rating, as seen in Figure A7. Having a high number of reviews does not necessarily suggest quality. As a whole, the bar types that find the most success tend to be those that combine food, atmosphere, and service in a cohesive and well-executed experience. On the other hand, bars that rely extensively on aesthetic or conceptual gimmicks may have a risk of disappointing customers despite initial appeal.

**Geospatial Visualization of Bar Opportunity Zones using Tableau**

To address the question of where to open a new bar in Philadelphia, we performed a geospatial analysis using tract-level aggregation and interactive filtering in Tableau.

We first identified "opportunity zones" - census tracts with:

- High average star ratings (above 4.2)
- Low bar density (fewer than 4 existing bars)

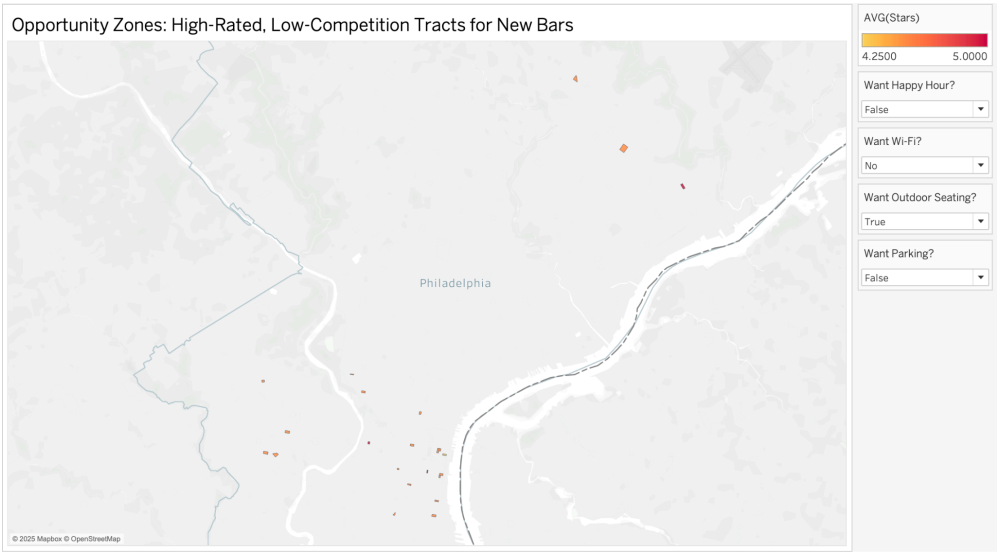
These zones represent areas with high customer satisfaction and low competition, suggesting strategic potential for new entrants.

We then developed an interactive Tableau dashboard that allows prospective bar owners to simulate their business plans. Users can select bar features such as Wi-Fi, outdoor seating, parking, and happy hour, and the map updates in real-time to show only tracts where those features are currently missing.

For example, a user can filter for locations where bars are highly rated but lack outdoor seating and parking, offering a way to fill that local gap. The ability to explore feature gaps empowers users to align business decisions with unmet local demand.

Additionally, we created a dynamic parameterized bar chart showing how specific features correlate with average ratings (Figure B4). This visualization highlights that bars offering outdoor seating or Wi-Fi tend to have slightly higher average ratings, reinforcing their importance in customer satisfaction.

We also identified hidden gem zones, or tracts where bars are highly rated (above 4.5) but have relatively few reviews. These neighborhoods may represent underserved or up-and-coming areas that are gaining popularity but haven't yet reached saturation.




**Conclusions**

Topic modeling insights revealed that the best-performing bars are those that have a strong, clear culinary identity combined with consistent service and ambiance. Bars with themes surrounding specific cuisines such as sushi or vegan options, as well as those that focus on quality ingredients and a welcoming atmosphere, consistently obtained positive ratings. On the other hand, bars that relied heavily on aesthetics or novelty such as rooftop setups or tech-heavy ordering typically underdelivered on customer satisfaction. Therefore, a future bar owner should consider opening a food-forward establishment that looks into both service quality and a coherent experience. A sushi-focused bar with good lighting, commendable service, and beautiful décor, for example, may be far more successful than a concept bar relying solely on flash or gimmicks.

The Tableau visualizations offer actionable insights for prospective bar owners by pinpointing census tracts in Philadelphia with high average ratings but low bar density, ideal zones for new openings. By layering in filters for features like outdoor seating, Wi-Fi, parking, and happy hour, the tool highlights neighborhoods where those amenities are currently lacking. For example, south and northeast tracts surfaced as strong opportunities: bars there are well-rated but often lack outdoor-friendly or tech-enabled environments. This empowers decision-makers to not only select *where* to open a bar, but also *what kind* of bar to launch, aligning business concepts with local gaps in the market.

Overall, the topic modeling and Tableau insights suggest that success lies not only in selecting a strong location but also in having a bar concept that is high in quality, consistent, as well as relevant to neighborhood needs.

## Appendix

**161385360554578. (2025, March 31). *Beloved gastropub files for bankruptcy again after closing 17 locations*. The US Sun.**
**https://www.the-sun.com/money/13907962/bar-louie-bankruptcy-second-time-mass-closures/**

**Figure A1. Distribution of Average Star Rating Scores Across All Topics**

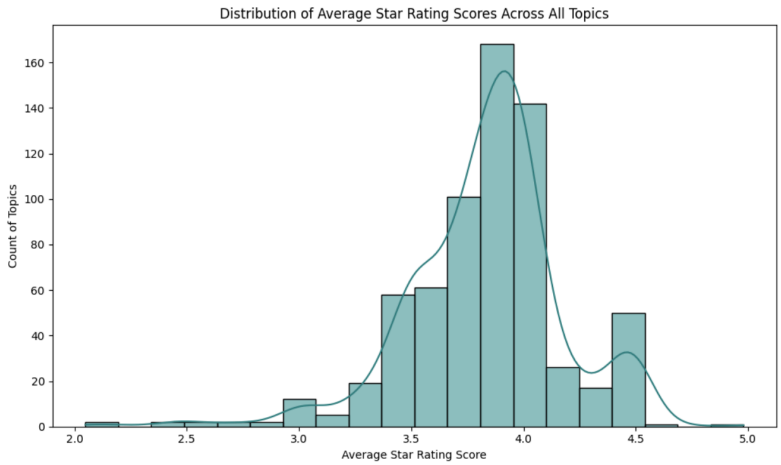



**Figure A2. Top 15 Most Frequent Topics**

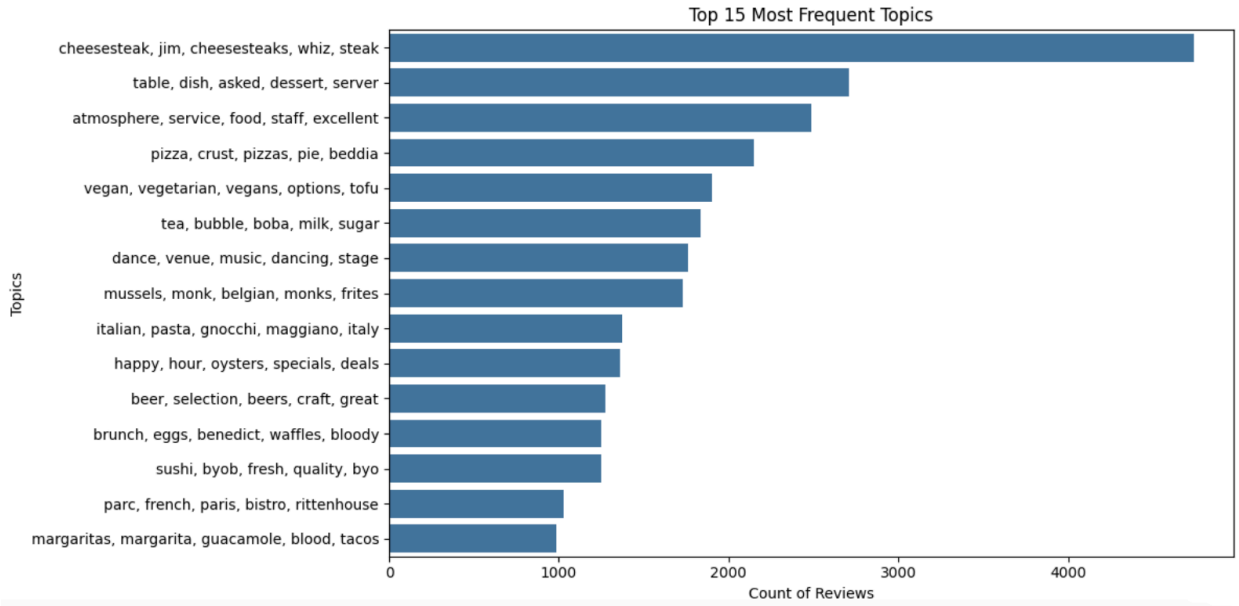


**Figure A3. Top 15 Topics by Frequency (with Average Rating Scores)**

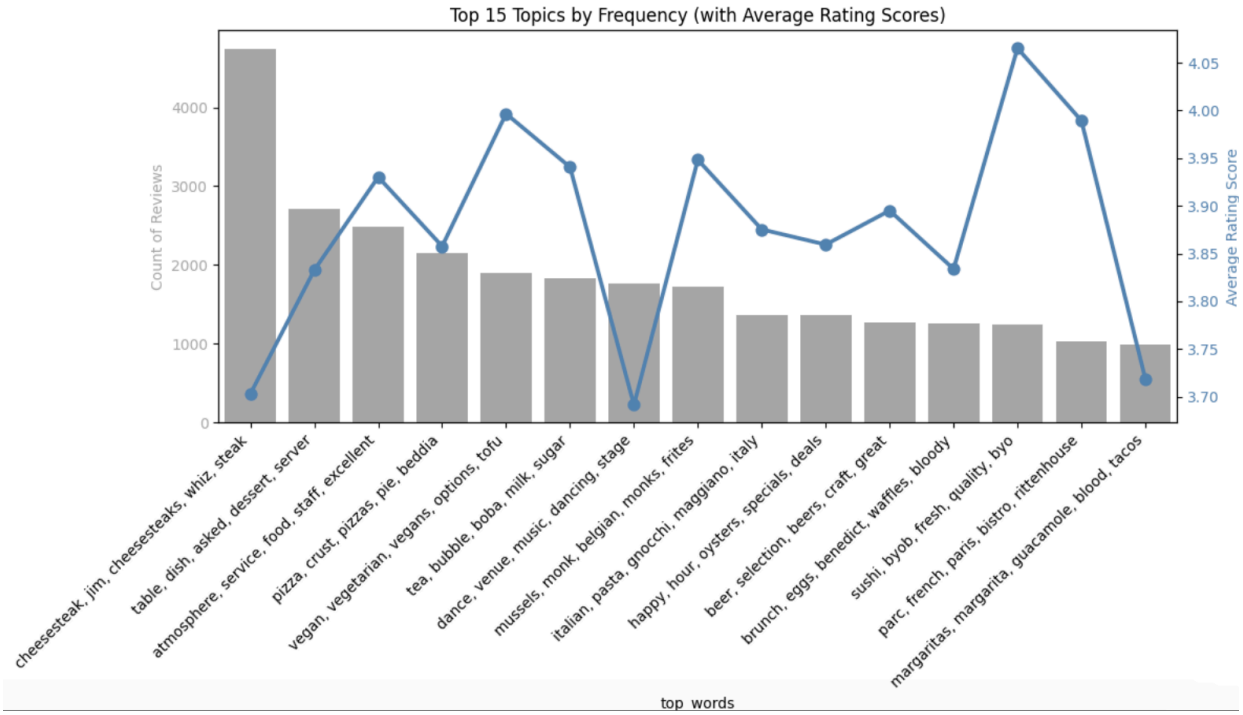



**Figure A4. Rating Distribution for Top 15 Topics (By Frequency)**

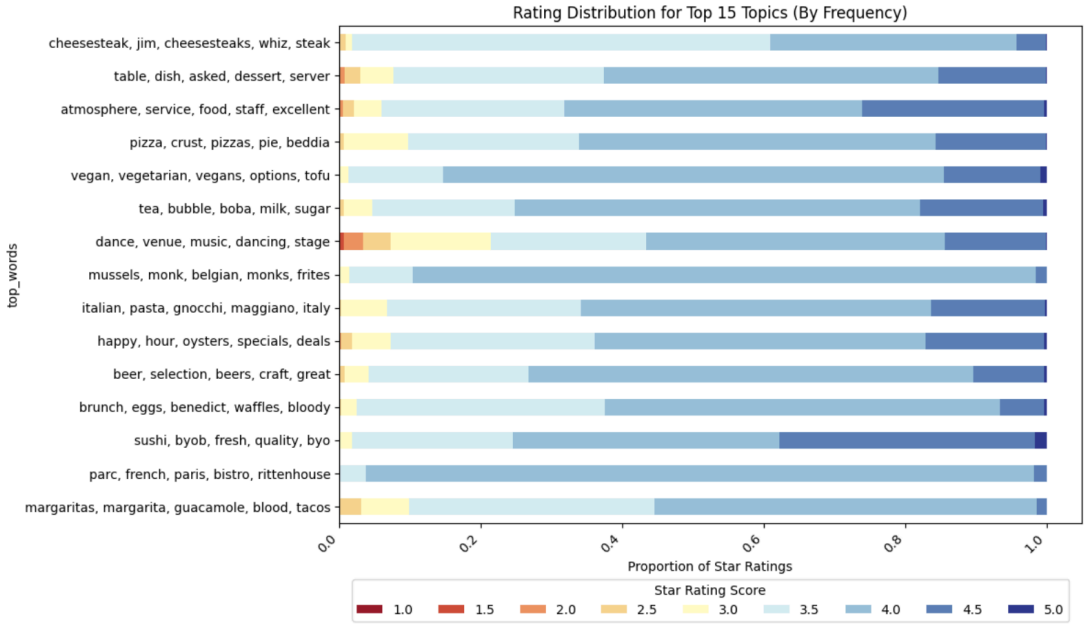


**Figure A5. Top 15 Topics by Average Rating (with Review Frequency)**

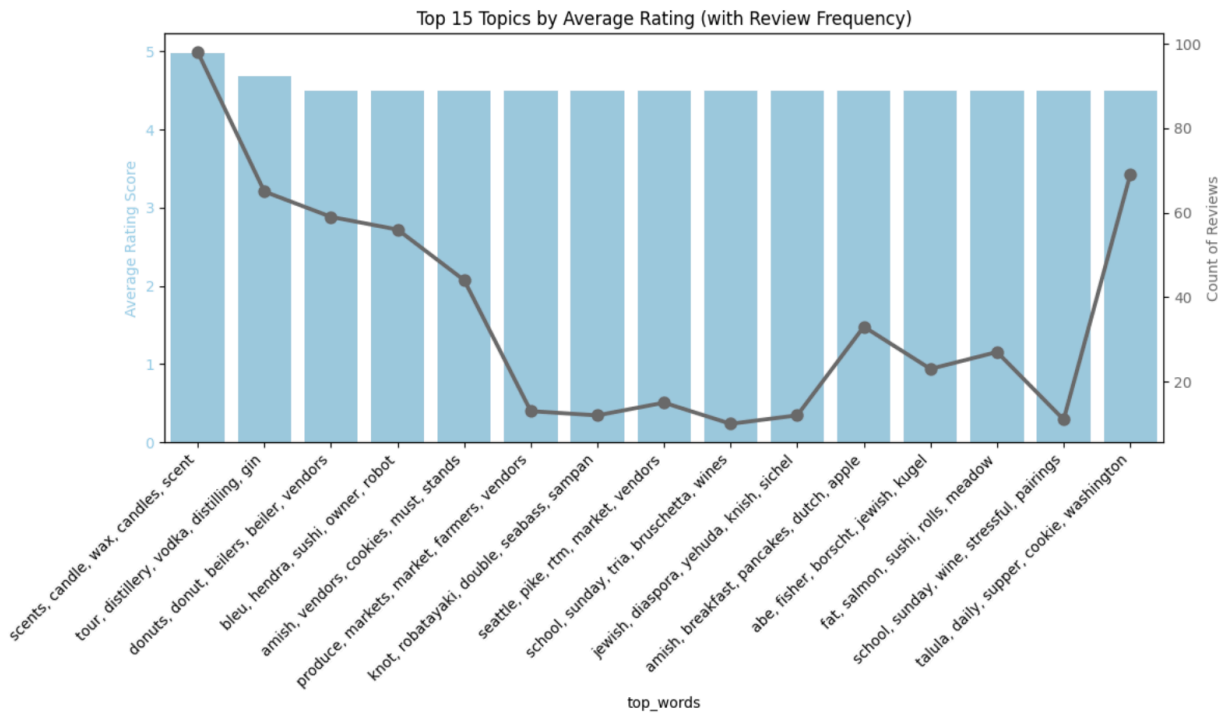



**Figure A6. Bottom 15 Topics by Average Rating (with Review Frequency)**

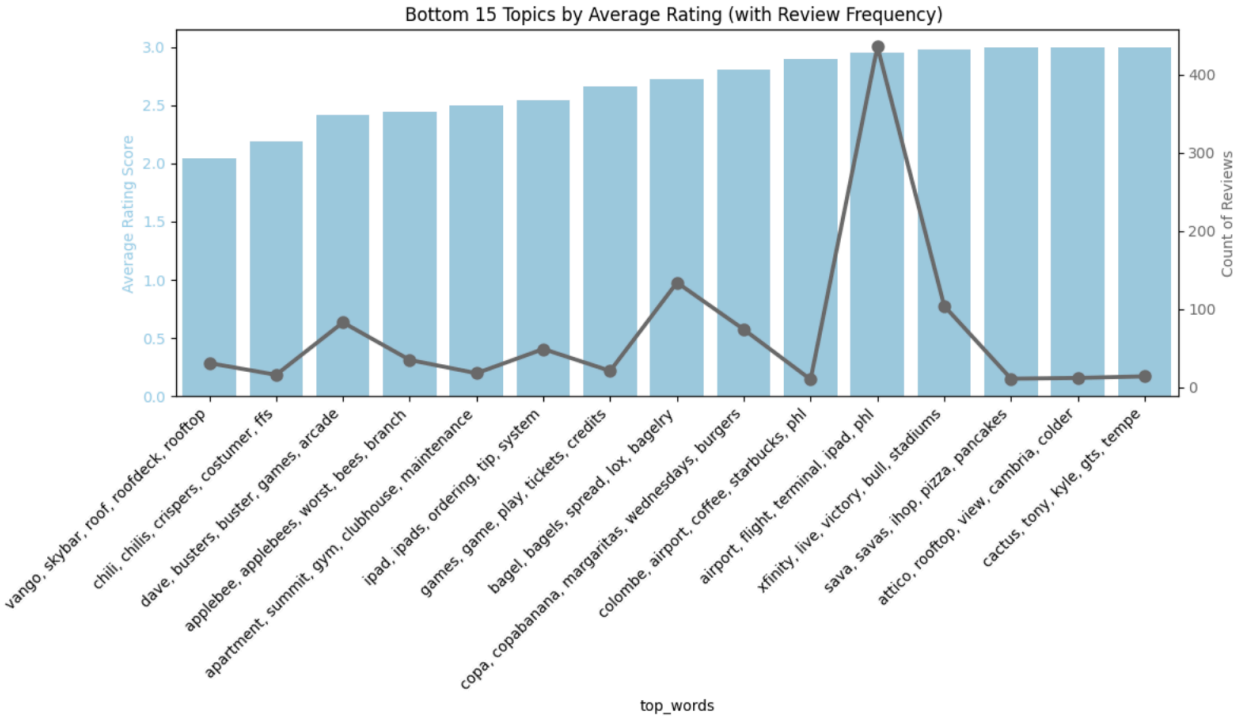


**Figure A7. Topic Frequency vs. Average Star Rating Scatterplot**

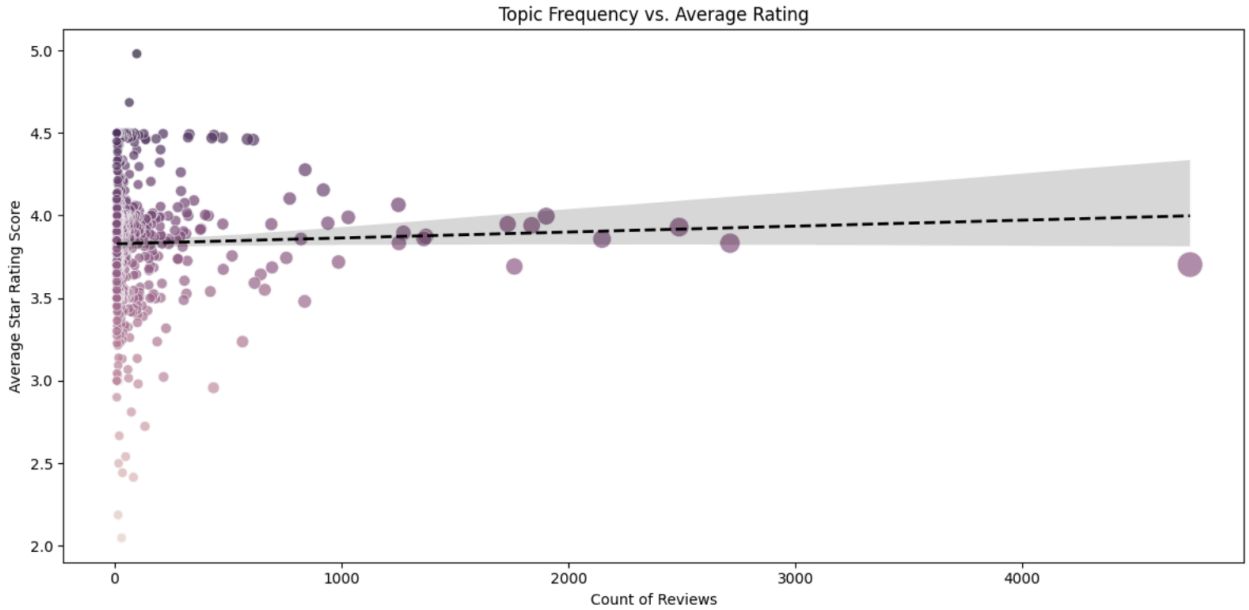



**Figure B1. A map displaying the average star rating of bars in each Geo ID**

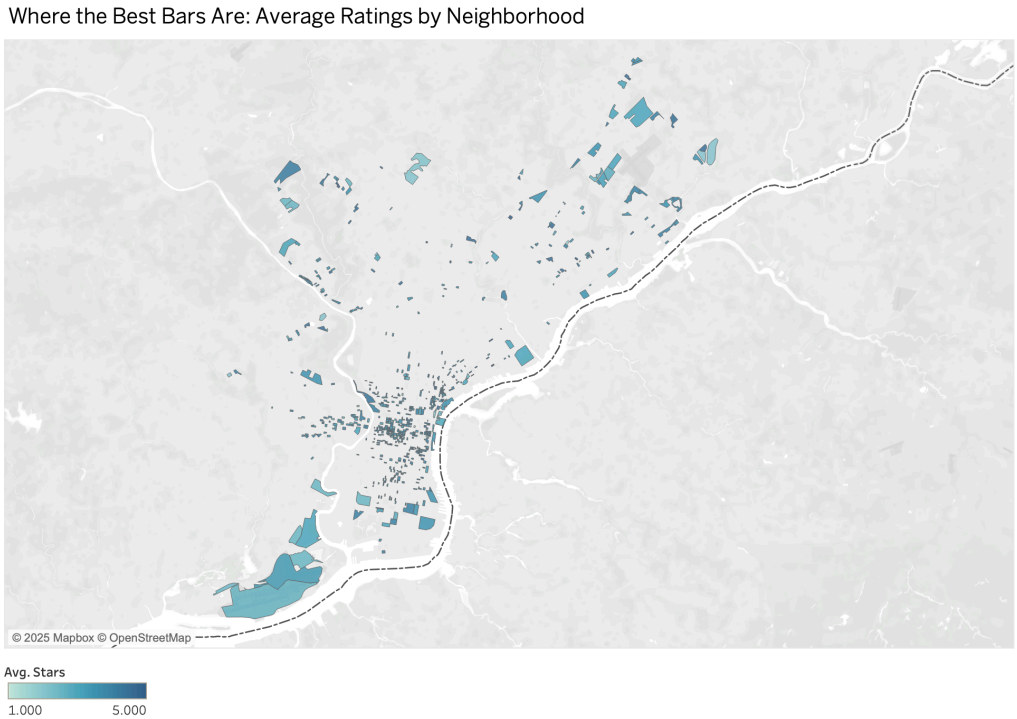


**Figure B2. A map showing the density of bars**

Where the Bars Are: A Census Tract Breakdown

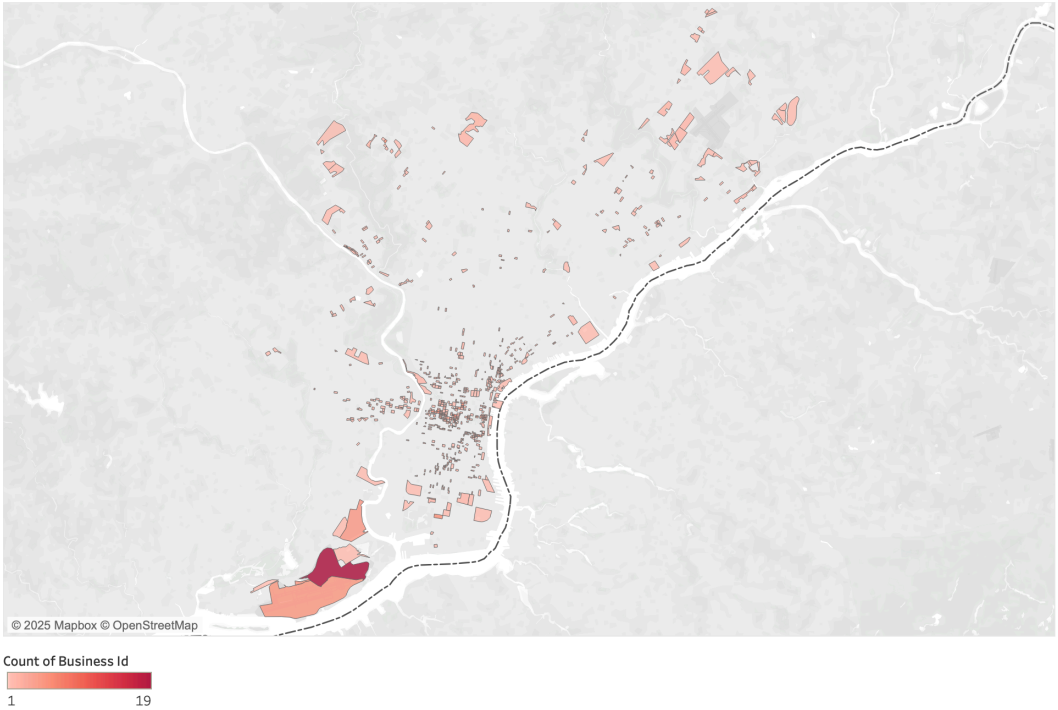
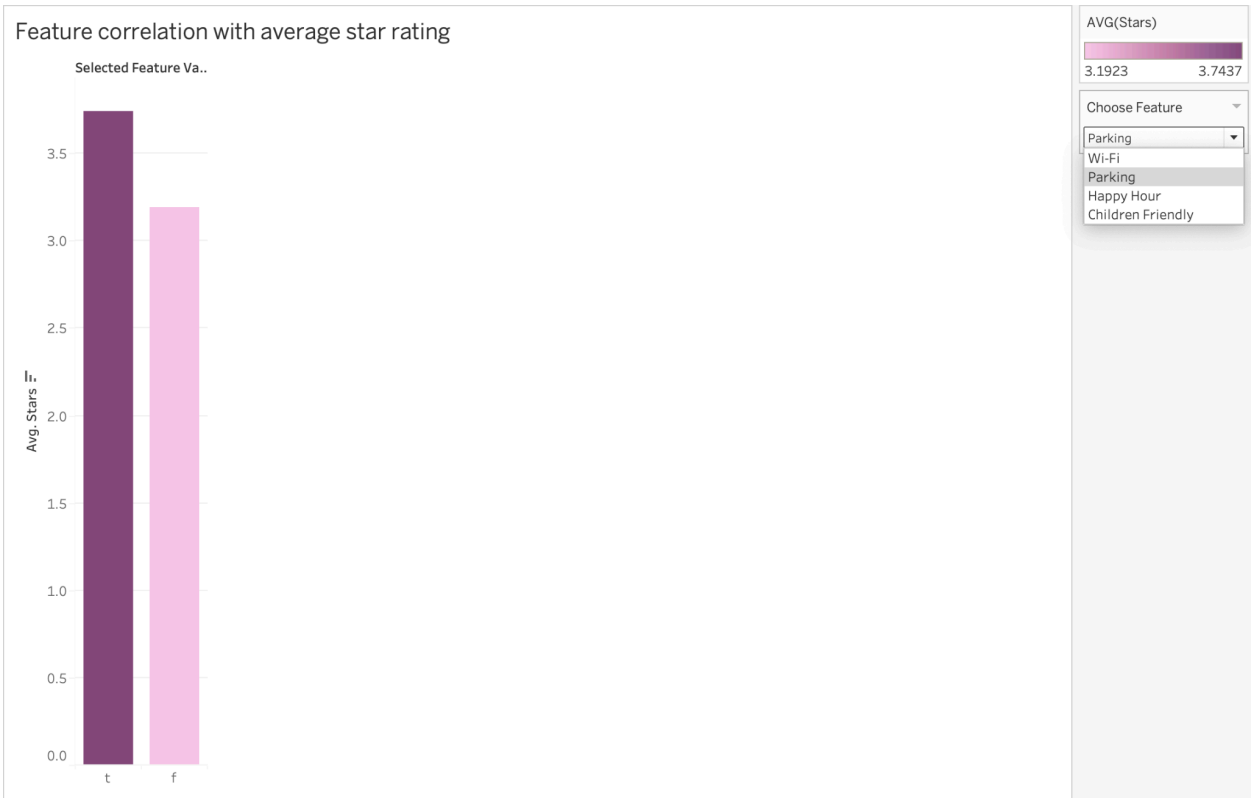



**Figure B3. A map highlights census tracts containing bars that are both highly rated (average stars above 4.5) and under the radar (fewer than 100 reviews)**

Hidden Gems: Top-Rated Bars with Low Visibility by Census Tract

**Figure B4. Feature correlation with average star rating**



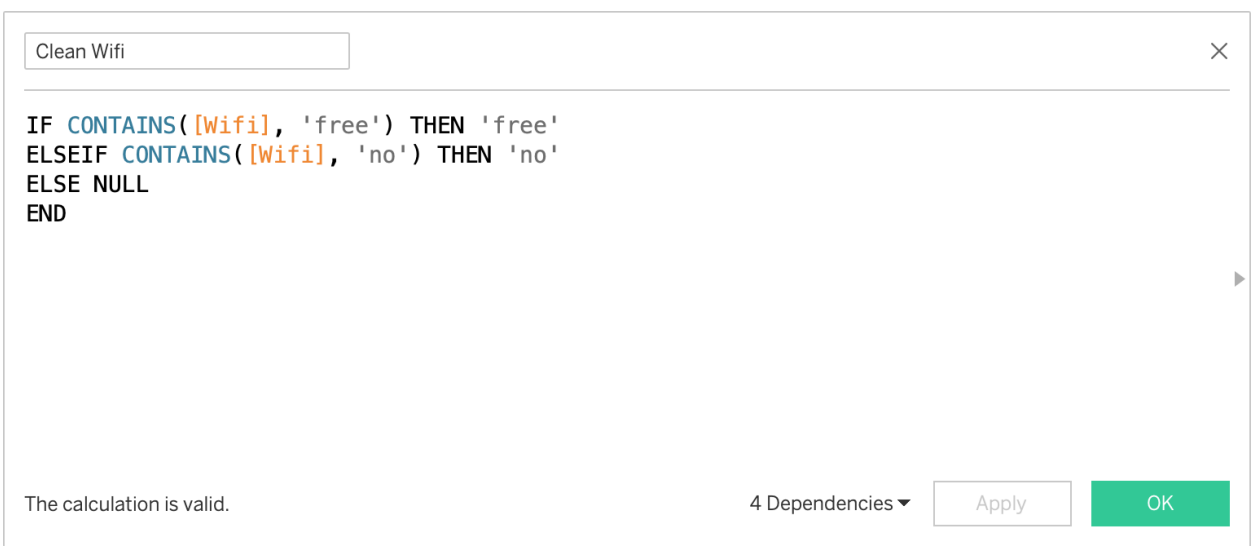


**Figure B5. An example of data cleaning in Tableau**